

Alternative form 2

What we derived from VLM is the total aero-coefficient derivatives with respect to the aircraft nose point ($x=0$).

$$C_{M\alpha}, C_{Z\alpha} \quad (x=0)$$

We know that the definition of NP is $\frac{\partial C_{M,np}}{\partial \alpha} = 0$

By moment translation equation, we move the total moment from NP to the nose

$$C_{M,0} = C_{M,np} - h_{np} \cdot C_Z$$

h is the distance to the nose, different from x_{np} .

The moment can be divided by the source

$$C_{M,0} = C_{M,0}^{WH} + C_{M,0}^{fus}$$

$\Rightarrow$ derivatives for both equations

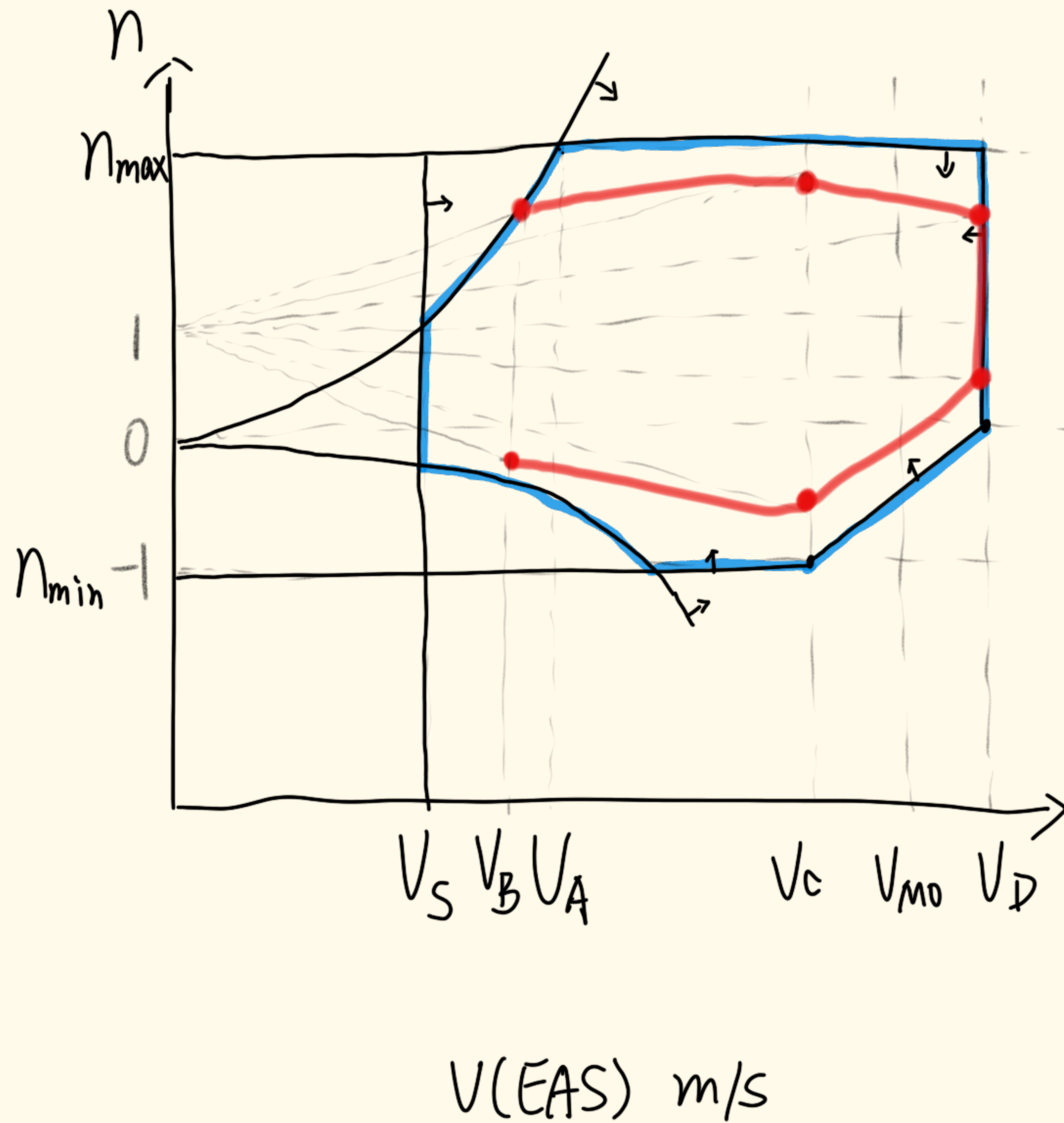
$$0 - h_{np} \cdot C_{Z\alpha} = C_{M\alpha}^{WH} + \frac{\partial C_M^{fus}}{\partial \alpha}$$

$$h_{np} = - \left(\frac{C_{M\alpha}^{WH}}{C_{Z\alpha}} + \frac{\partial C_M^{fus}}{\partial \alpha} \cdot \frac{\partial \alpha}{\partial C_Z} \right) \quad (C_Z^{fus} \approx 0)$$

$$= - \left(\frac{C_{M\alpha}^{WH}}{C_{Z\alpha}} + \frac{\partial C_M^{fus}}{\partial C_Z} \right) \quad \leftarrow \text{in code}$$

$$= - \left(\frac{\partial C_M^{WH}}{\partial C_Z} + \frac{\partial C_M^{fus}}{\partial C_Z} \right) \quad \leftarrow (14)$$

$n = L/W$ $v - EAS$ the dynamic pressure won't change with altitude



n_{max} : max limit load

$$\begin{cases} +2.5 & , \geq 50,000 \text{ lbs} \\ \min(2.1 + \frac{24000}{W+10000}, 3.8) & , \leq 50,000 \text{ lbs} \end{cases}$$

n_{min} : min limit load
-1

V_C : the design EAS

V_{Mo} : the maximum operating EAS

$$V_{Mo} = 1.06 V_C$$

V_D : the design diving EAS

$$V_D = 1.07 V_{Mo}$$

$$n = \frac{L}{W} = \frac{\frac{1}{2} \rho V^2 S C_L}{W} = \frac{1}{2} \rho_{SL} V_{EAS}^2 \frac{C_L}{(W/S)}$$

$$n(V_{EAS}) \leftarrow \rho_{SL} \cdot W/S \cdot C_L$$

V_A : the design maneuvering speed

$$V_A = \sqrt{\frac{2n_{\max}(W/S)}{C_{L,\max} \rho_{SL}}}$$

V_S : stall speed at normal level flight

$$V_S = \sqrt{\frac{2(W/S)}{C_{L,\max} \rho_{SL}}}$$

$$n_g = 1 + \left[\frac{\rho_{SL}}{2} \frac{1}{(W/S)} C_L^2 U \right] V_{EAS} \quad U - \text{gust vertical velocity}$$

$$n_g(V_{EAS}) \leftarrow \rho_{SL} \cdot \frac{W}{S} \cdot C_L^2 \cdot U$$

$U_{e,c}$ the regulated gust velocity at cruise speed
 $U_{e,D}$ the regulated gust velocity at diving speed
 $U_{e,B}$ rough gust velocity

} from FAR 25

V_B : design speed for maximum gust intensity

$$\frac{\rho_{SL} V_B^2 C_{Lmax}}{2(W/S)} = 1 + \frac{\rho_{SL} C_L^{\alpha} U_{eB} V_B}{2(W/S)} \Rightarrow V_B$$

(grey for understanding, black for writing)